

ARUN KUMAR A N

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [WhatsApp](#)

PROFILE SUMMARY

Full Stack Developer with hands-on experience in enterprise software development and full-stack application development using **React, TypeScript, NestJS, MySQL, and REST APIs**. Experienced in developing modular backend services, scalable application workflows, asynchronous processing, database-driven applications, and responsive user interfaces. Strong understanding of software architecture, API development, debugging, system-level workflows, authentication, containerized development, and Git-based development practices. Experienced in designing loosely coupled application components with a strong focus on scalability, reliability, maintainability, and clean code.

CORE TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, React.js, TypeScript, TanStack Query v5, React Hook Form, Zod, Tailwind CSS

Backend & APIs: Node.js, NestJS, RESTful APIs, TypeORM, JWT Authentication

Databases & Queues: MySQL 8, Redis (Caching & Bull Queues), Relational Database Design

Storage & Cloud: MinIO Object Storage

Testing & Tools: Postman, Swagger, GitHub

PROFESSIONAL WORK EXPERIENCE

Inventech Info Solutions Pvt. Ltd.

Bengaluru, India

Full Stack Developer Intern

Dec 2025 – Present

- Developed full-stack web applications using **React, TypeScript, NestJS, REST APIs, and MySQL**.
- Designed modular backend services and implemented RESTful APIs for application workflows and frontend integration.
- Worked on database design, API integration, authentication, validation, debugging, and application performance improvements.
- Developed reusable React components and responsive user interfaces while maintaining structured frontend-backend communication.
- Applied Git-based development workflows, API testing, debugging, and clean coding practices throughout application development.

ENTERPRISE PLATFORMS ENGINEERED AT INVENTECH

Layout Master – Enterprise Layout Planning & Administration System

- Designed Database Architecture in **MySQL & TypeORM** creating 5 Master Tables: **Persons (Employees), Posts (Designations), Roles, Permissions (Create, View, Update, Delete), and Menus (Navigation items)**.
- Architected 4 Mapping Tables to connect the system: **Role-to-Post Mapping, Post-to-Person Mapping** (with start/end date tenure), **Menu-to-Role Mapping** (sidebar menus), and **Permission-to-Role Mapping** (action authorization).
- Developed **NestJS REST APIs** using Controllers, DTOs with validation (**class-validator**), and TypeORM Services to handle CRUD operations and cascading relationships across all mapping tables.
- Engineered **Dynamic Menu & Action Authorization Engine**, where a user's active Post and Role determine what menus render in the React sidebar and which buttons (Add, Edit, Delete) are enabled or hidden.
- Built Full-Stack Admin Screens in **React, TypeScript, and TanStack Query**, allowing administrators to easily assign employees to posts, map menus to roles, and toggle permissions with real-time UI table updates.
- **Tech Stack:** React, TypeScript, Vite, TanStack Query, Tailwind CSS, NestJS, TypeORM, MySQL, JWT

KEONICS – Karnataka State Electronics Development Corporation (Tenant Space & Procurement ERP)

- **Workflow JSON Schema & Process Graph:** Designed structured JSON process graphs (**processmaster.json**) capturing **taskId, process_id, task_type** (STARTTASK, NORMAL, DECISION, ENDTASK), and branching **decision_flows** (Approve, Clarification, Reject) to govern application lifecycles.
- **Database Schema & Instance Tracking:** Architected TypeORM entities and custom repositories (**process_instance, task_instance, task_history**) in MySQL to track active workflow pointers, task assignments, and lifecycle states.
- **Role-Based State Machine & Task Mapping:** Implemented a backend state transition engine using a **TaskToRole** lookup dictionary to dynamically map each workflow task to authorized roles (Estate Officer, Zone Commissioner, Commissioner).
- **Validation & Action Payloads:** Engineered RESTful action endpoints (**POST /send-for-approval, PUT /close-task/:id**) using NestJS DTOs and class-validator, where officers submit decisions, mandatory audit comments, and verification data.

- **Traceable Audit History:** Logged every transition into a history table containing previous state, new state, acting actor ID (Actor A or Actor B), remarks, and timestamps for compliance auditing.
- **Tech Stack:** React, TypeScript, Ant Design, Redux Toolkit, NestJS, TypeORM, MySQL, AWS S3

IEBA – Workforce & Personal Time Study SaaS Platform

- **Assigned Program & Activity Resolution Engine:** Developed backend APIs to dynamically fetch and filter only the assigned programs and active industrial activities mapped to an employee's department role, ensuring accurate logging.
- **Personal Time Study (PTS) Logging & Validation:** Engineered the core time-entry APIs (`/user/timeentry/record`, `/submit`) with strict DTO validation that records activity start/end times and total minutes while preventing overlapping time intervals.
- **Supervisor Review & Approval Pipeline:** Built the supervisor review workflow (`/timestudyrecord/:id/review`) enabling managers to fetch their reportees' submitted weekly time studies, inspect activity breakdowns, and approve or reject them with feedback.
- **Scheduled Time Study (STS) Configuration:** Architected the scheduled time study module allowing administrators to create time study schedules, assign specific eligible users, and restrict logging strictly within the scheduled time windows.
- **Backend Services & Frontend API Integration:** Implemented modular NestJS controllers, TypeORM services, and relational MySQL schemas, seamlessly integrating with the React frontend to synchronize employee time logs and supervisor approval queues.
- **Tech Stack:** React, TypeScript, Vite, TanStack Query, Tailwind CSS, NestJS, TypeORM, MySQL, JWT

WorkspHERE – Organization Timesheet & Attendance Management System

- Built a 3-tier organizational hierarchy portal (Admin, Manager, Employee) managing weekly timesheet logging, multi-level manager approvals, and delegate workflows.
- Configured **Redis-backed Bull Queues** for automated weekly reminder dispatches, submission deadline alerts, and email notifications via Nodemailer.
- Implemented interactive productivity and attendance analytics dashboards using **ApexCharts** with real-time aggregated metrics.
- Automated one-click PDF timesheet generation and exports using **jsPDF** and **autoTable** with custom styling.
- **Tech Stack:** React, TypeScript, Redux Toolkit, Ant Design, ApexCharts, NestJS, Bull Queues, Redis, MySQL, TypeORM

CDRMS – Correct Dimension Report Management System (PWD Land Survey)

- Developed cross-platform mobile survey application in **React Native (Expo SDK)** supporting offline survey drafts for field civil engineers.
- Implemented real-time GPS geotagging, on-site photo/video evidence capture, and local caching for low-connectivity environments.
- Built secure review and verification endpoints backed by **NestJS transactions and MySQL** for Khatedar land record verification.
- Structured role-based access control and approval flows connecting field surveyors, reviewing officers, and administrative authorities.
- **Tech Stack:** React Native, Expo, TypeScript, React Navigation, gluestack-ui, NestJS, MySQL, TypeORM

CAS / CSA – Civic Amenity Site Allotment & Citizen Registration Portal

- Developed end-to-end citizen registration and commercial CA-site allotment modules with multi-step registration, form validation (**Zod + React Hook Form**), and refund processing.
- Integrated multilingual localization support using **react-i18next** alongside secure SMS-based OTP authentication for citizens and administrative staff.
- Built automated digital certificate generation pipelines and secure document uploads backed by **AWS S3**.
- Engineered high-concurrency **NestJS REST APIs** backed by TypeORM query builders, MySQL, and Redis caching.
- **Tech Stack:** React, TypeScript, Ant Design, Redux Toolkit, TanStack Query, NestJS, MySQL, Redis, AWS S3

INDEPENDENT PROJECTS & OPEN-SOURCE RESEARCH

Voice Excel Assistant – AI Spreadsheet Client & Backend API

[Frontend GitHub](#) | [Backend GitHub](#)

- Built a hands-free spreadsheet client leveraging the **OpenAI Whisper API** and **Web Speech API** for voice-driven cell data entry with column-aware data parsing.
- Engineered an auto-saving grid engine with 500ms debounce persistence, row-level locking, and state management.
- Integrated real-time **ExcelJS streaming** for high-speed .xlsx file generation and immediate exports.
- Structured modular Node.js/Express backend APIs for audio transcript ingestion, formatting, and file streaming.
- **Tech Stack:** React, TypeScript, OpenAI Whisper, Web Speech API, Node.js, Express, ExcelJS

Smart Healthcare System – AI Symptom Analysis & Scheduling (MCA Project)

[GitHub](#)

- Developed an intelligent telemedicine portal featuring AI-guided symptom checking and automated doctor specialty recommendation.
- Built an appointment booking calendar with real-time slot management, patient medical record vaults, and role-based access control.

- Implemented secure authentication with JWT, protected patient data encryption, and doctor prescription management.
- Designed responsive, user-friendly clinical dashboards for patients, doctors, and hospital administrators.
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js (MERN), Tailwind CSS, JWT

Interactive School & Academy Dashboard (Inventech Project)

[GitHub](#)

- Designed a responsive timetable scheduling interface using **React Big Calendar** and Lucide icons.
- Built comprehensive analytics dashboards in **Recharts** displaying student attendance trends and examination metrics.
- Implemented form validation with React Hook Form and Zod for course and student enrollment records.
- **Tech Stack:** React.js, TypeScript, Recharts, React Big Calendar, Tailwind CSS, Zod

Realtime Chat & Messaging Platform

[GitHub](#)

- Engineered a bidirectional WebSocket event architecture using **Socket.IO** for low-latency live messaging.
- Implemented active online user rosters, typing indicator triggers, and message delivery status updates.
- Structured persistent conversation history storage in MongoDB with indexed queries for instant retrieval.
- **Tech Stack:** Node.js, Express.js, Socket.IO, React, MongoDB

Online Multi-Vehicle Rental System (BCA Final Semester Project)

[GitHub](#)

- Built multi-category vehicle catalog filtering by class, pricing, and availability status.
- Engineered reservation booking workflow with automated rental duration calculations and billing breakdowns.
- Integrated automated PDF invoice generation and administrative fleet tracking panels.
- **Tech Stack:** PHP, MySQL, HTML5, CSS3, JavaScript, FPDF

Developer Portfolio

[GitHub](#) | [Live Demo](#)

- Designed and built a modern, production-ready developer portfolio showcasing projects, technical skills, and work experience.
- Integrated smooth animations, contact form handling via EmailJS, and responsive dark/light theme.
- Deployed on Vercel with optimized asset delivery and continuous deployment from GitHub.
- **Tech Stack:** React.js, TypeScript, Tailwind CSS, TanStack Query, EmailJS, Vercel

Pharmacy Management System

[GitHub](#)

- Built application to manage medical inventory, point-of-sale billing, and supplier records.
- Implemented automated low-stock notifications, expiration date alerts, and sales reporting.
- **Tech Stack:** Python / Java, MySQL, Relational Database Design

API TESTING & TOOLING

Postman: Conducted API testing by configuring automated test assertions (`pm.test`) for HTTP status codes, schema validation, and payload integrity across environments with pre-request JWT auth scripts.

Swagger: Implemented interactive Swagger UI for NestJS REST APIs using decorators (`@ApiProperty`, `@ApiOperation`) for live endpoint testing and client-server contract verification.

EDUCATION

Master of Computer Applications (MCA)	2024 – 2025
<i>Maharaja Institute of Technology, Mysore</i>	CGPA: 8.5 / 10.0
Bachelor of Computer Applications (BCA)	2021 – 2023
<i>St. Joseph's First Grade College, Mysore</i>	CGPA: 8.4 / 10.0

LANGUAGES KNOWN

English (Professional), Kannada (Native)